

```
import pandas as pd

df = pd.read_csv("/content/heart_disease_uci.csv")
df.head()
```

	id	age	sex	dataset	cp	trestbps	chol	fb	restecg	thalch	exang	oldpeak	slope	ca	thal	num
0	1	63	Male	Cleveland	typical angina	145.0	233.0	True	lv hypertrophy	150.0	False	2.3	downsloping	0.0	fixed defect	0
1	2	67	Male	Cleveland	asymptomatic	160.0	286.0	False	lv hypertrophy	108.0	True	1.5	flat	3.0	normal	2
2	3	67	Male	Cleveland	asymptomatic	120.0	229.0	False	lv hypertrophy	129.0	True	2.6	flat	2.0	reversable defect	1
3	4	37	Male	Cleveland	non-anginal	130.0	250.0	False	normal	187.0	False	3.5	downsloping	0.0	normal	0
4	5	41	Female	Cleveland	atypical angina	130.0	204.0	False	lv hypertrophy	172.0	False	1.4	upsloping	0.0	normal	0

```
# Check missing values
print(df.isnull().sum())

# Fill numeric missing values with median
df = df.fillna(df.median(numeric_only=True))

# Encode categorical columns (if any)
df = pd.get_dummies(df, drop_first=True)

df.head()
```

```

id          0
age         0
sex         0
dataset     0
cp          0
trestbps    59
chol        30
fbs         90
restecg     2
thalch      55
exang       55
oldpeak     62
slope      309
ca         611
thal       486
num         0
dtype: int64

```

	id	age	trestbps	chol	thalch	oldpeak	ca	num	sex_Male	dataset_Hungary	...	cp_non-anginal	cp_typical_angina	fbs_True	restecg_normal	restecg_st-t abnormality	exang_True	slope_flat
0	1	63	145.0	233.0	150.0	2.3	0.0	0	True	False	...	False	True	True	False	False	False	False
1	2	67	160.0	286.0	108.0	1.5	3.0	2	True	False	...	False	False	False	False	False	True	True
2	3	67	120.0	229.0	129.0	2.6	2.0	1	True	False	...	False	False	False	False	False	True	True
3	4	37	130.0	250.0	187.0	3.5	0.0	0	True	False	...	True	False	False	True	False	False	False
4	5	41	130.0	204.0	172.0	1.4	0.0	0	False	False	...	False	False	False	False	False	False	False

5 rows × 23 columns

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# Change target column name if needed
y = df["num"] # If error, check column name using df.columns
X = df.drop("num", axis=1)

```

```

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

```

```

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

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```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

k_values = [3, 5, 7, 11, 15]

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results = {}
f1_scores = []

for k in k_values:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train_scaled, y_train)

    y_pred = knn.predict(X_test_scaled)

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, average='weighted')
    rec = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')

    results[k] = (acc, prec, rec, f1)
    f1_scores.append(f1)

    print(f"---- k = {k} ----")
    print("Accuracy :", acc)
    print("Precision:", prec)
    print("Recall   :", rec)
    print("F1 Score :", f1)
    print()

```

```

---- k = 3 ----
Accuracy : 0.5217391304347826
Precision: 0.4648880676142484
Recall   : 0.5217391304347826
F1 Score : 0.4843954287503416

```

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---- k = 5 ----
Accuracy : 0.532608695652174
Precision: 0.48826758531489406
Recall   : 0.532608695652174
F1 Score : 0.5060527287120297

```

```

---- k = 7 ----
Accuracy : 0.5489130434782609
Precision: 0.5086800802848636
Recall   : 0.5489130434782609
F1 Score : 0.5235698821237574

```

```

---- k = 11 ----
Accuracy : 0.5652173913043478
Precision: 0.5258235122365559
Recall   : 0.5652173913043478
F1 Score : 0.5398709185487238

```

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---- k = 15 ----
Accuracy : 0.5597826086956522
Precision: 0.5121052272938564
Recall   : 0.5597826086956522
F1 Score : 0.529732406775547

```

```

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))

```

```
import pandas as pd

results_df = pd.DataFrame(results,
                           index=["Accuracy", "Precision", "Recall", "F1-score"]).T

results_df
```

	Accuracy	Precision	Recall	F1-score
3	0.521739	0.464888	0.521739	0.484395
5	0.532609	0.488268	0.532609	0.506053
7	0.548913	0.508680	0.548913	0.523570
11	0.565217	0.525824	0.565217	0.539871
15	0.559783	0.512105	0.559783	0.529732

```
import matplotlib.pyplot as plt

plt.plot(k_values, f1_scores, marker='o')
plt.xlabel("k value")
plt.ylabel("F1 Score")
plt.title("k vs F1 Score")
plt.show()
```

